



TB Drug-Resistance Decision Support

A neuro-symbolic clinical reasoning system for *Mycobacterium tuberculosis*, validated against real-world resistance data

■ Why a hybrid design

Each layer answers a different clinical need, and together they make the reasoning both auditable and adaptable.

1

Auditable rules

A symbolic rule engine encodes WHO resistance logic, so every classification can be traced to the rule that produced it.

2

Case experience

Case-based reasoning draws on prior patient cases for the situations where guidelines alone do not fix a regimen.

3

Open knowledge

A knowledge graph and a plain-English query layer make the WHO catalog explorable without writing code.

■ Four-layer architecture

Knowledge graph

The WHO catalog as 30,699 mutation nodes

Rule engine

Classifies MDR, pre-XDR, and XDR, then selects the regimen

Case-based reasoning

Retrieval over 1,000 synthetic patient cases

Natural-language interface

Plain English to Cypher under a read-only guard

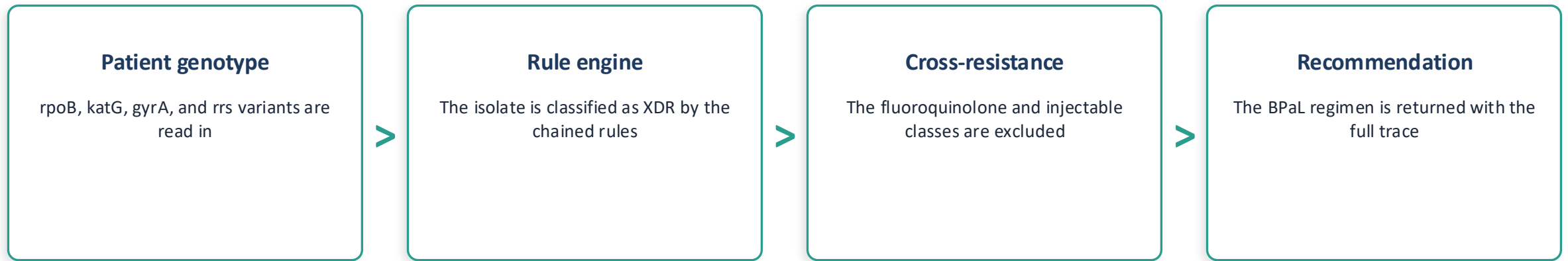
Durable platform

The knowledge graph and rule engine are grounded in the WHO catalog. This layer is stable and evidence-based, and it carries the clinical logic.

Swappable patient layer

The case base sits on top and can be replaced with real cohorts as they become available, without disturbing the platform.

■ The system in action



Every step is explicit, so a clinician can see why a regimen was chosen and what was ruled out.

■ The symbolic core works

The rule engine classifies resistance tiers from catalogued mutations, applies whole-class cross-resistance so a contraindicated drug is never offered, and selects between the BPaL and BPaLM regimens.

Its logic is explicit and testable rather than learned, which is what makes each recommendation defensible to a clinician.

100%

of the twelve scored natural-language
queries answered correctly

■ Grounded in the WHO catalog

30,699

mutation nodes in the knowledge graph

48,152

rows of the WHO mutation catalog

1,000

transparent synthetic patient cases

The platform is grounded in current WHO evidence, while the patient layer stays transparent and deterministic given a fixed seed.

■ Validated on real-world isolates

The rule engine was scored against measured drug-susceptibility phenotypes from the CRyPTIC consortium.

13,118

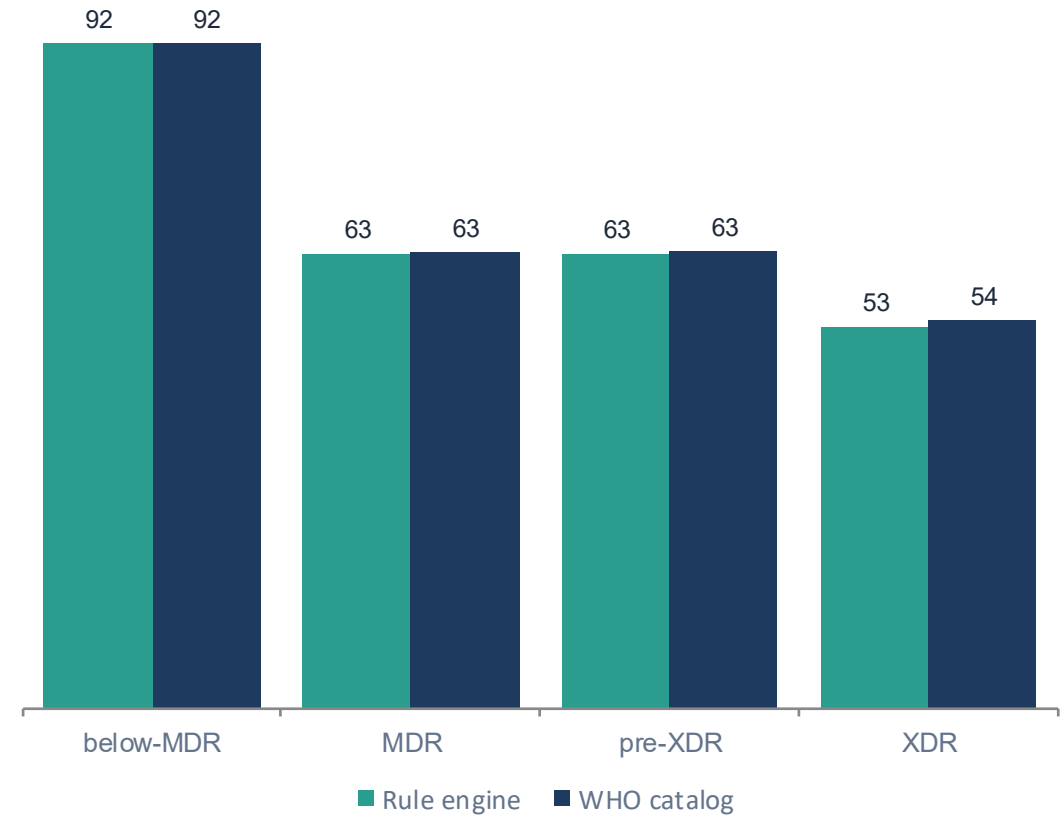
real measured isolates

99.9%

agreement with the WHO catalog

Reproducing the gold-standard catalog on real isolates shows the reasoning is faithful, not approximate.

Accuracy by tier, engine versus catalog



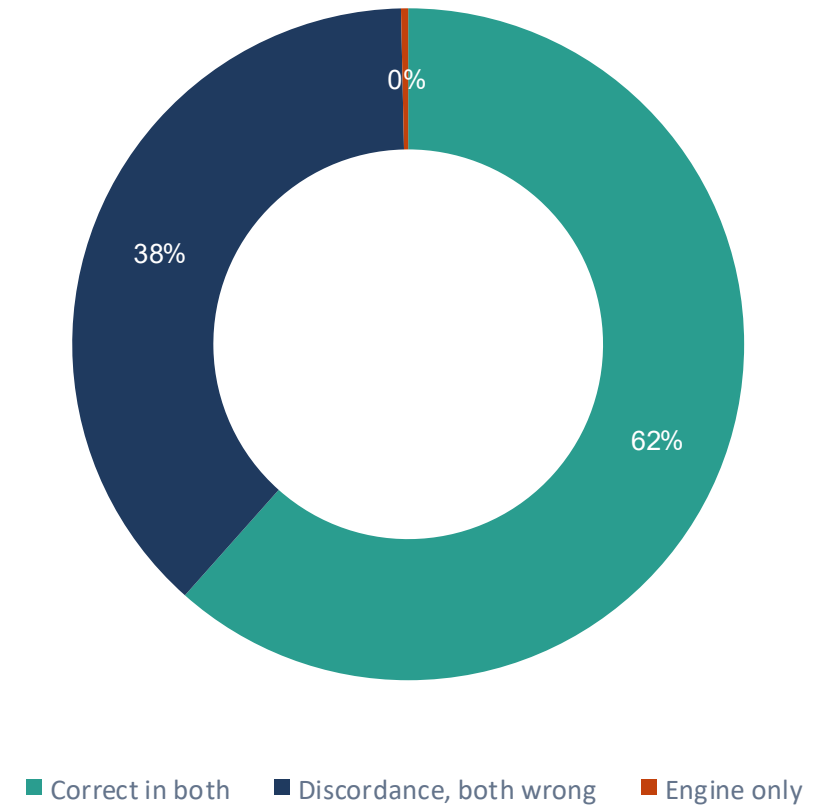
■ The ceiling is biological, not a defect

Of 3,505 resistant isolates, almost every error the engine makes is also made by the gold-standard catalog.

Those shared errors are genotype-phenotype discordance, resistance seen on testing with no catalogued genotypic marker. No genotype-based method can recover them, so the limit is biological.

12 errors unique to the engine, against **1,335** shared with the catalog

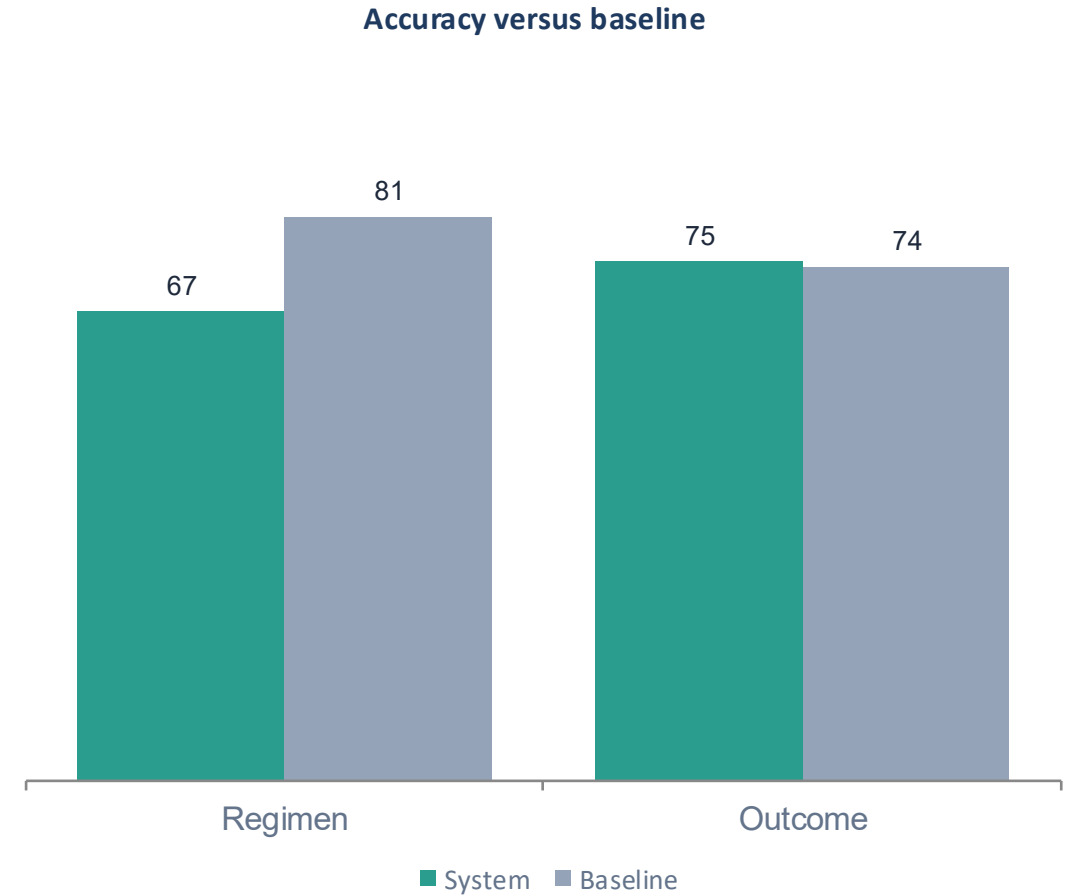
Resistant isolates, 3,505 total



■ Case retrieval, the experimental component

The case-based layer is reported honestly as the experimental component.

Regimen accuracy sits below the majority baseline. The shortfall is diagnosed as objective mismatch plus retrieval starvation in the rare resistant classes, where neighbor-based retrieval is data-poor by construction, rather than left unexplained.



■ Engineering rigor

32 deterministic tests

A single pytest suite locks in classification, calibration math, generator determinism, and graph integrity, with no database or API required.

Calibration tested and rejected

Expected calibration error is 0.075 on the raw probability. Temperature scaling was tested and rejected on evidence, since it raised the error to 0.177.

Reproducible evaluation

One entry point produces every metric, the graph arm is skipped gracefully when services are down, and a fixed seed makes runs repeatable.

■ Limitations

Synthetic patient layer

No real outcome dataset with the needed structure was available, so case retrieval is shown on generated cases.

No outcomes in CRyPTIC

The real dataset carries genotype and phenotype only, so it validates classification, not regimen outcomes.

Scoped XDR definition

The engine implements the pre-2021 definition as a deliberate, documented choice.

Discordance ceiling

About 16% of measured resistance has no catalogued genotypic marker, so no rule-based method can reach it.

■ Future work

A learned model

Train on the full genome-wide variant table and minimum-inhibitory-concentration magnitudes, to probe how much of the genotype-phenotype discordance can be recovered beyond the curated catalog. The catalog defines the bar the model must beat.

Outcome validation

Evaluate the case-based layer against the TB Portals dataset, which carries real treatment outcomes, so the regimen and outcome layer can finally be measured on real patients rather than synthetic cases.